Q4 — Reflection and next steps (1 page)

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What the traces showed.

- **The router over-clarified (v2).** It never sees the data, so the rule "no task data → ask" fired on every status question, and routing fell to 74%. Telling the router that the specialists have all the data, and adding 8 routing examples, fixed it (v3: 100%).
- **Naive prompts misused real data.** Facts computed in code kept every ID real, but v1 adopted a user's 30% goal and named owners in a client email.
- **Two wrong answers passed loose checks (v3).** T10 dropped the Blocked count, and T5 reported project health as Sprint 3's. I found them only by reading every answer. The prompt fixed T10 and E5; T5 needed per-sprint facts computed in code.
- **Variance.** Three runs of v5 showed T1 dropping phase names 1 time in 3; a one-line Planner rule fixed it (v6: 69/69). A fifth agent for schedule questions (v7) passed 75/75 over 3 runs.
- **Lesson:** a green test suite is only as good as its assertions, and one run is not evidence.

Improvement plan (next two weeks).

1. A check that summary rows add up to the total.
2. CSV upload in chat, with column validation.
3. A PM-set reporting date instead of a fixed one.
4. Hierarchical routing for two-part requests (E10).
5. A Monday Milestone Alert digest to Slack, approved by the PM.
6. An LLM-as-judge score in Langfuse for tone and completeness, calibrated against human review.

Evaluation and fine-tuning.

- **The ground truth is a regression suite.** Every prompt version is promoted only after 3 live runs score at least as well as the current one. This is how a QA team gates builds.
- **Fine-tuning comes later, and only if prompts stop improving scores.** The training data would be PM-corrected outputs from the pilot's edit feedback, plus fixed failing traces: about 100–200 examples for a gpt-4o-mini fine-tune, judged on the same suite. Until then, prompts plus code give more for less.

Privacy. Client project data goes to OpenAI's API (no training on API data) only after security and legal sign-off.

- Each agent sees only the data it needs.
- Owner names are personal data under GDPR: client emails exclude them, and production should replace them with placeholders in prompts.
- Langfuse stores inputs and outputs. Tracing currently uses the US region; before the pilot it moves to the EU region with a retention limit.
- API keys live in n8n credentials, never in the exported workflow.

- Prompt-injection attempts are ignored (E3), and nothing is ever auto-sent.

Rollout for 10 PMs/TPMs.

- **Week 0, readiness:** sign-off, training and feedback buttons.
- **Weeks 1–2, champions:** 3 PMs on the ABCDE project. Gate: ≥ 70% of outputs sent with minor or no edits, and 0 fabricated facts reaching a client.
- **Weeks 3–4, pilot:** all 10 PMs, 2+ projects. Gate: 8 of 10 weekly active, status report under 15 minutes.
- **Week 5:** the CTO decides go, adjust or stop.
- **Change management:** start with the weekly status report, keep the PM as author, and show sources on every answer. See the program charter.